Lista 04: Integrais e Aplicações

1. Resolva as integrais indefinidas abaixo:

1) $\int 10\left(\sqrt[3]{x^2}\right)dx$

2) $\int \frac{2}{\sqrt[3]{x}}dx$

3) $\int y^3(2y^2-3)dy$

4) $\int (8x^4+4x^3-6x^2-4x+5)dx$

5) $\int (\frac{3x^3}{5}-4x^2+\frac{5x}{2}-3)dx$

6) $\int (2x^3-\frac{5}{3x}+4x-6)dx$

7) $\int \left(3\sqrt{x}-\frac{4}{3x^2}\right)dx$

8) $\int \sqrt{x}(x+1)dx$

9) $\int \left(\frac{2}{x^3}+\frac{3}{x^2}+5\right)dx$

10) $\int \frac{y^4+2y^2-1}{\sqrt{y}}dy$

11) $\int \sqrt[3]{3x-4}dx$

12) $\int x^2(x^3-1)^{10}dx$

13) $\int 5x\left(\sqrt[3]{(9-4x^2)^2}\right)dx$

14) $\int x\left(\sqrt{x+2}\right)dx$

15) $\int \frac{x^3}{\sqrt{1-x^2}}dx$

16) $\int \frac{e^{2x}}{\sqrt{1-e^x}}dx$

17) $\int x(2x^2+3)^{10}dx$

18) $\int \frac{(1+\sqrt{x})^3}{\sqrt{x}}dx$

19) $\int \frac{t-2}{(t^2-4t+3)^3}dt$

20) $\int \sqrt{1-4y}dy$

21) $\int \sqrt[3]{6-2x}dx$

22) $\int x\sqrt{x^2-9}dx$

23) $\int x^2(x^3-1)^{10}dx$

24) $\int \frac{y^3dy}{(1-2y^4)^5}dy$

25) $\int (x^2-2x+1)^{5/2}dx$

26) $\int \frac{x(3x^2+1)dx}{(3x^4+2x^2+1)^2}$

27) $\int x^5(x^3-1)^{10}dx$

28) $\int \cos x(2+\operatorname{sen} x)^5 dx$

29) $\int \frac{4\operatorname{sen} x}{(1+\cos x)^2}dx$

30) $\int sen(3x)dx$

2) Calcular as integrais definidas abaixo:

a) $\int_0^3 (3x^2 - 4x + 1)dx$

b) $\int_0^6 (x^2 - 2x)dx$

c) $\int_1^2 \frac{x^2 + 1}{x^2} dx$

d) $\int_0^1 \frac{z}{(z^2 + 1)^3} dz$

e) $\int_1^{10} \sqrt{5x - 1} dx$

f) $\int_{-2}^0 3w\sqrt{4 - w^2} dw$

g) $\int_0^1 \frac{x^2}{(x^3 + 1)^5} dx$

3) Use a integral definida para encontrar a área limitada pelas curvas abaixo. Faça um esboço da região.

a) $y = 4 - x^2$ e o eixo x.

b) $y = x^2$, retas $x = 0$, $y = 0$ e $x = 3$.

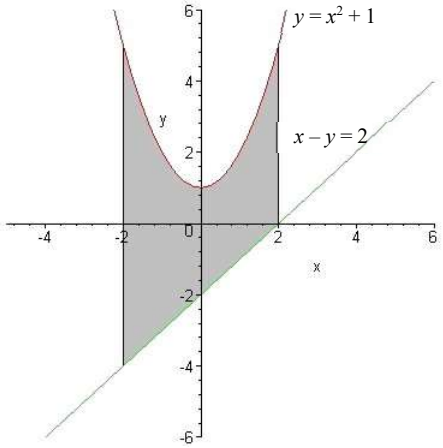
c) $y = x^2 - 7$ e $y = 1 - x^2$.

d) $y = -x^2 + 5x - 4$ e $y = x - 1$.

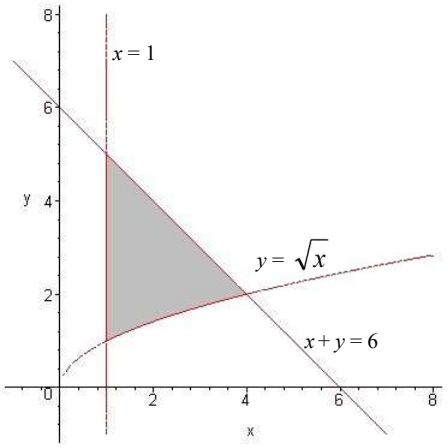
e) $y = x^2 - 6x + 5$ e o eixo x.

4. Estabeleça uma integral que possa ser usada para determinar a área da região sombreada e encontre o valor da área correspondente.

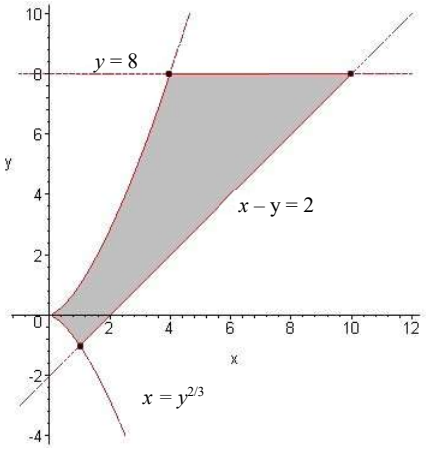
a)



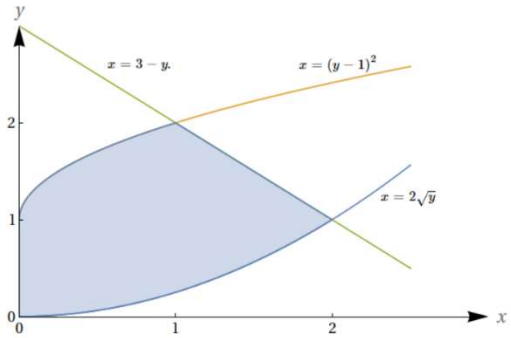
b)



c)



d)



Boa Sorte!

GABARITO – LISTA 04

1. Resolva as integrais indefinidas abaixo:

1) x

2) $3x^{2/3} + c$

3) $\frac{y^6}{3} - \frac{3y^4}{4} + c$

4) $\frac{8x^5}{5} + x^4 - 2x^3 - 2x^2 + 5x + c$

5) X

6) X

7) X

8) $\frac{2x^{5/2}}{5} + \frac{2x^{3/2}}{3} + c$

9) $-\frac{1}{x^2} - \frac{3}{x} + 5x + c$

10) x

11) x

12) $\frac{1}{33}\left(x^3 - 1\right)^{11} + c$

13) $-\frac{3}{8}\left(9 - 4x^2\right)^{5/3} + c$

14) x

15) $\frac{e^x}{x+1} + c$

16) $-x^2\sqrt{1-x^2} - \frac{2}{3}(1-x^2)^{3/2} + c$

17) x

18) $\frac{1}{44}(2x^2+3)^{11} + C$

19) $\frac{1}{2}(1+\sqrt{x})^4 + C$

20) $-\frac{1}{4(t^2-4t+3)^2} + C$

21) $-\frac{1}{6}(1-4y)^{3/2} + C$

22) $-\frac{3}{8}(6-2x)^{4/3} + C$

23) $\frac{1}{3}(x^2-9)^{3/2} + C$

24) $\frac{1}{33}(x^3-1)^{11} + C$

25) $\frac{1}{32(1-2y^4)^4} + C$

26) $\frac{3}{11}(x-2)^{11/3} + C$

27) $-\frac{1}{4(3x^4+2x^2+1)} + C$

28) $\frac{1}{6}\left(2 + \operatorname{sen} x\right)^6 + c$

2) Calcular as integrais definidas abaixo:

a) 12 b) 36 c) $\frac{3}{2}$ d) $\frac{3}{16}$

e) $\frac{134}{3}$ f) -8 g) $\frac{5}{64}$

3) Use a integral definida para encontrar a área limitada pelas curvas abaixo:

a) 36
b) 9
c) $\frac{5}{64}$
d) ***
e) ***

4. Estabeleça uma integral que possa ser usada para determinar a área da região sombreada e encontre o valor da área correspondente.